

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)	(i)		Space based clearer / detect more types of radiation (that don't penetrate atmosphere / no interference from Earth's atmosphere / no interference from light pollution	1			1		
		(ii)		Earth based are cheaper / hard to maintain / hard to repair in space	1			1		
	(b)			years		1		1		
	(c)	(i)		X – rays	1			1		
		(ii)		Radio (waves)	1			1		
		(iii)		Any 2 × (1) from: Gamma (rays) (1) u-v / ultra-violet (1) i-r/infra-red (1) micro(waves) (1) Do not accept ultra or violet Do not accept infra or red X-rays, visible light and radio waves neutral	2			2		
	(d)			Selection of 2 m (1) Ans = 150 000 000 Hz (1) Ans of 150 000 000 only (2) Ans of 30 000 000 (1 only) Accept 50 000 000 (1)		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			Big Bang	1			1		
				Question 5 total	7	3		10	2	